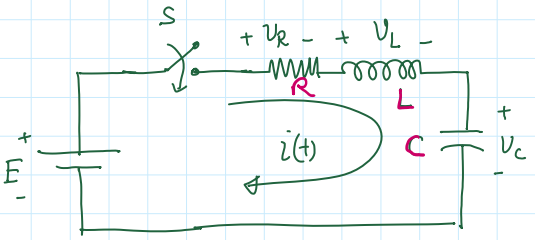
Lecture 11C: Step Response to RLC Series Circuit

Consider the following circuit:



$$v_R + v_L + v_C = E \quad \cdots (1)$$

$$iR + L\frac{di}{dt} + \frac{1}{C}\int_{-\infty}^{t} i\,dt = E$$

$$iR + L\frac{di}{dt} + \underbrace{\frac{1}{C}\int_{-\infty}^{0} i\,dt}_{V_{C0}} + \frac{1}{C}\int_{0}^{t} i\,dt = E$$

$$\Rightarrow iR + L\frac{di}{dt} + \frac{1}{C}\int i\,dt = E - V_{C0}$$

$$\text{or} \quad L\frac{d^2i}{dt^2} + R\frac{di}{dt} + \frac{i}{C} = 0$$

$$\text{or} \quad \frac{d^2i}{dt^2} + \frac{R}{L}\frac{di}{dt} + \frac{i}{LC} = 0 \quad \cdots (*)$$

A second order differential equation that need to be solved for $i(t)$.

Let $\frac{d}{dt}$ = Differential operator = $S = D$

$$\left(S^2 + \frac{R}{L}S + \frac{1}{LC}\right) I(s) = 0$$

$$\boxed{S^2 + \frac{R}{L}S + \frac{1}{LC} = 0} \Rightarrow \text{Characteristic equation of the system}$$

$$S_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-\frac{R}{L} \pm \sqrt{\left(\frac{R}{L}\right)^2 - \frac{4}{LC}}}{2}$$

$$S_{1,2} = -\frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$$

$$i(t) = K_1 e^{S_1 t} + K_2 e^{S_2 t}$$

Depending upon the value of $\left(\frac{R}{2L}\right)^2$ compared to $\frac{1}{LC}$, the following nature of roots are present.

(A) $\left(\frac{R}{2L}\right)^2 > \frac{1}{LC}$, two distinct real roots

Ⓐ $\left(\frac{R}{2L}\right)^2 > \frac{1}{LC}$, two distinct real roots

Overdamped response

Ⓑ $\left(\frac{R}{2L}\right)^2 = \frac{1}{LC}$, two equal real roots

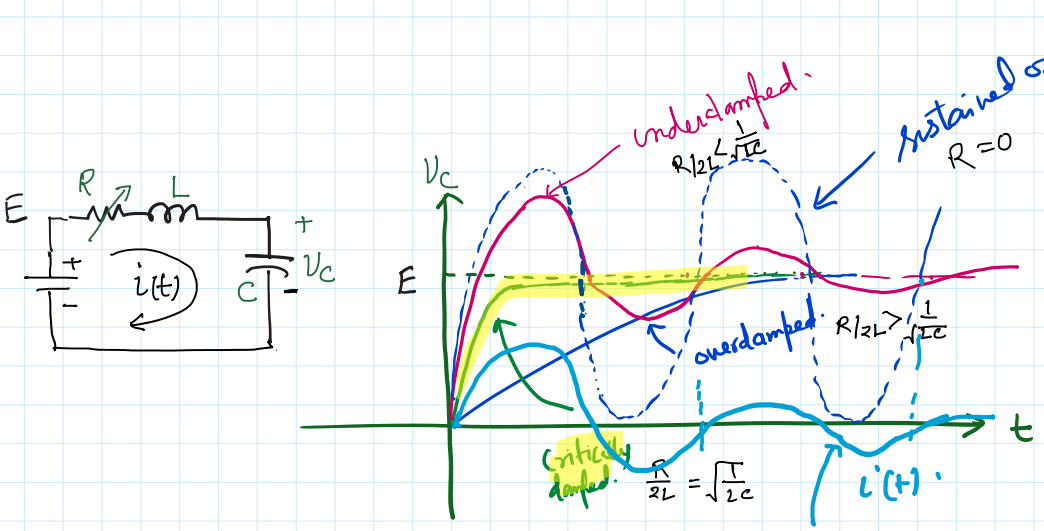
Critically damped

Ⓒ $\left(\frac{R}{2L}\right)^2 < \frac{1}{LC}$; complex conjugate roots $\quad s_{12} = -\frac{R}{2} \pm j\sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2}$

Underdamped system

Ⓓ $R = 0 \quad s_{1,2} = \pm j\sqrt{\frac{1}{LC}}$

Roots are imaginary; Sustained oscillation



In the context of this, few important terms are defined.

① Natural frequency: $\omega_n = \frac{1}{\sqrt{LC}}$ rad/sec.

② Critical Resistance (R_c): $\left.\frac{R}{2L}\right|_{R=R_c} = \frac{1}{\sqrt{LC}}$

$R_c = 2\sqrt{\frac{L}{C}}$

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$R > R_c \Rightarrow$ overdamp

$R < R_c \Rightarrow$ underdamp

③ Damping Ratio (ζ) $= \frac{R}{R_c} = \frac{R}{2}\sqrt{\frac{C}{L}}$

$$\boxed{\zeta = \frac{R}{2}\sqrt{\frac{C}{L}}}$$

④ Damped frequency $\omega_d = \omega_n\sqrt{1-\zeta^2}$

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$$\zeta = \frac{R}{2}\sqrt{\frac{C}{L}} = \frac{1}{2}\frac{R}{L}\sqrt{\frac{C}{L}}\,L = \frac{1}{2}\frac{R}{L}\cdot\sqrt{LC}$$

$$s^2 + \frac{R}{L}s + \frac{1}{LC} = 0$$

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

$$\zeta = \frac{R}{L}\cdot\frac{1}{2\omega_n}$$

$$s_{1,2} = \frac{-2\zeta\omega_n \pm \sqrt{(2\zeta\omega_n)^2 - 4\omega_n^2}}{2}$$

$$\frac{R}{L} = 2\zeta\omega_n \quad (\#)$$

$$\frac{R}{2L} = \zeta\omega_n \quad (\#)$$

$$s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$

$$\boxed{s_{1,2} = -\zeta\omega_n \pm j\omega_d}$$

$$i(t) = K_1 e^{s_1 t} + K_2 e^{s_2 t} = K_1 e^{(-\zeta\omega_n + j\omega_d)t} + K_2 e^{(-\zeta\omega_n - j\omega_d)t}$$

$$= e^{-\zeta\omega_n t}\left[K_1 e^{j\omega_d t} + K_2 e^{-j\omega_d t}\right]$$

$$= e^{-\zeta\omega_n t}\left[K_1\cos\omega_d t + jK_1\sin\omega_d t + K_2\cos\omega_d t - jK_2\sin\omega_d t\right]$$

$$= e^{-\zeta\omega_n t}\left[(K_1+K_2)\cos\omega_d t + j(K_1-K_2)\sin\omega_d t\right]$$

$$\boxed{i(t) = e^{-\zeta\omega_n t}\left(K_1'\cos\omega_d t + K_2'\sin\omega_d t\right)}$$

K_1' and K_2' with $i(0)$ and $\frac{di(0)}{dt}$

Given $i(0) = 0$

$$0 = 1\,(K_1' + K_2' \times 0) \Rightarrow K_1' = 0$$

$$\boxed{i(t) = e^{-\zeta\omega_n t}\cdot K_2'\sin\omega_d t} \quad (\#)$$

$$R\,i^{\,0} + L\frac{di(0)}{dt} + \frac{1}{C}\int_0^t i\,dt = (E - V_{C0})$$

$$\left.\frac{di}{dt}\right|_{t=0} = \left(\frac{E - V_{C0}}{L}\right)$$

$$\frac{di}{dt} = e^{-\zeta\omega_n t}\cdot 1\,(K_2'\cos\omega_d t\,\omega_d) + e^{-\zeta\omega_n t}(-\zeta\omega_n)\cdot K_2\sin\omega_d t$$

$$K_2' = \frac{di/dt\,|_{t=0}}{\omega_d}$$

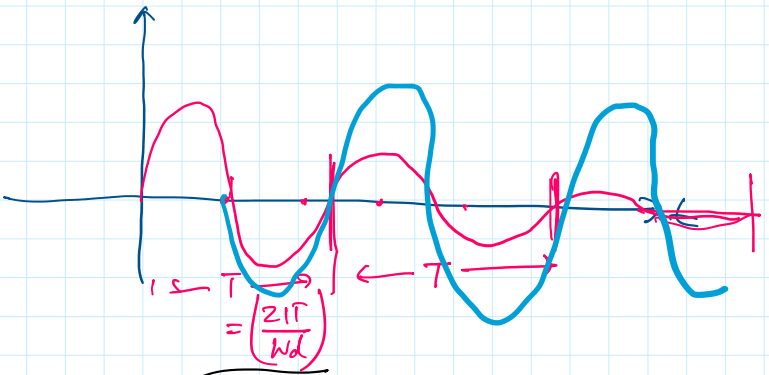
dt

$$K_2' = \frac{di/dt\,|_{t=0}}{\omega_d}$$

$$K_2' = \frac{E - V_{C0}}{\omega_d L}$$

$K_2 \sin \omega_d t$

$$i(t) = \left(\frac{E - V_{C0}}{\omega_d L}\right) e^{-\zeta \omega_n t} * \sin \omega_d t$$



$$T = 2\pi f_d = \omega_d$$

$$\frac{2\pi}{T_d} = \omega_d$$

$$T_d = \frac{2\pi}{\omega_d}$$

$$= \left(\frac{2\pi}{\omega_d}\right)$$

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}$$

$$\omega_d = \omega_n$$

$$V_R = iR = \frac{(E - V_{C0})}{\omega_d L} * \left(\frac{R}{L}\right) e^{-\zeta \omega_n t} \sin \omega_d t$$

$$\frac{(E - V_{C0})}{\omega_n \sqrt{1-\zeta^2}} \quad 2\zeta \omega_n e^{-\zeta \omega_n t} \sin \omega_d t$$

$$V_R = iR = \frac{E - V_{C0}}{\sqrt{1-\zeta^2}} \cdot 2\zeta e^{-\zeta \omega_n t} \sin \omega_d t \qquad V_R$$

$$V_L = L\frac{di}{dt} ? = \frac{d}{dt}\left(\frac{E - V_{C0}}{\omega_d} e^{-\zeta \omega_n t} \cdot \sin \omega_d t\right)$$

$$= \frac{E - V_{C0}}{\omega_d} \frac{d}{dt}\left(e^{-\zeta \omega_n t} \sin \omega_d t\right)$$

$$= \frac{E - V_{C0}}{\omega_d} \left[e^{-\zeta \omega_n t} (-\zeta \omega_n) \sin \omega_d t + e^{-\zeta \omega_n t} \cos \omega_d t \; \omega_d \right]$$

$$= \frac{E - V_{C0}}{\omega_n \sqrt{1-\zeta^2}} e^{-\zeta \omega_n t} \left[-\zeta \omega_n \sin \omega_d t + \omega_n \sqrt{1-\zeta^2} \cdot \cos \omega_d t \right]$$

$$= \frac{E - V_{C0}}{\sqrt{1-\zeta^2}} e^{-\zeta \omega_n t} \left[-\zeta \sin \omega_d t + \sqrt{1-\zeta^2} \cos \omega_d t \right]$$

$\gamma \cos\phi$ $\qquad$ $\gamma \sin\phi$

$$= -\frac{E - V_{C0}}{\sqrt{1-\zeta^2}} e^{-\zeta \omega_n t} \left[\zeta \sin \omega_d t - \sqrt{1-\zeta^2} \cos \omega_d t \right]$$

$\zeta = \gamma \cos\phi$

$\sqrt{1-\zeta^2}$

$$v_L = -\frac{E-V_{C0}}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_d t - \phi)$$

$\zeta = r\cos\phi$

$\sqrt{1-\zeta^2} = r\sin\phi$

$r = 1$

$\phi = \tan^{-1}\frac{\sqrt{1-\zeta^2}}{\zeta}$

$v_C = ?$ $\quad \frac{1}{C}\int i\,dt$

$$v_R + v_L + v_C = E$$

$$v_C = E - v_R - v_L = E - (v_R + v_L)$$

$$= E - \left\{ \frac{E-V_{C0}}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \left[2\zeta \sin\omega_d t - \zeta \sin\omega_d t + \sqrt{1-\zeta^2}\cos\omega_d t \right] \right\}$$

$$= E - \left[\frac{E-V_{C0}}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \left(\zeta \sin\omega_d t + \sqrt{1-\zeta^2}\cos\omega_d t \right) \right]$$

($\zeta = r\cos\phi$, $\sqrt{1-\zeta^2} = r\sin\phi$)

$$v_C(t) = E - \left\{ \frac{E-V_{C0}}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_d t + \phi) \right\}$$

$$v_C(t) = E\left[1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \sin(\omega_d t + \phi) \right]$$

$V_{C0} = 0$

$V_C = 0$